

```
In [4]: import pandas as pd
df=pd.DataFrame({
    'Patient_ID':['P001','P002','P003','P004','P005'],
    'Age':[45,52,38,61,49],
    'Tumor_Size':[2.1,3.5,1.8,4.2,2.9],
    'ER':[1,1,0,1,0],
    'HER2':[0,1,1,0,1],
    'Lymph_Node':[0,1,0,1,1]
})
print(df)
df.info()
df['ER']=df['ER'].astype('category')
df['HER2']=df['HER2'].astype('category')
df['Lymph_Node']=df['Lymph_Node'].astype('category')
df.describe()
```

	Patient_ID	Age	Tumor_Size	ER	HER2	Lymph_Node
0	P001	45	2.1	1	0	0
1	P002	52	3.5	1	1	1
2	P003	38	1.8	0	1	0
3	P004	61	4.2	1	0	1
4	P005	49	2.9	0	1	1

```
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Patient_ID  5 non-null     str
1   Age         5 non-null     int64
2   Tumor_Size  5 non-null     float64
3   ER          5 non-null     int64
4   HER2        5 non-null     int64
5   Lymph_Node  5 non-null     int64
dtypes: float64(1), int64(4), str(1)
memory usage: 372.0 bytes
```

```
Out[4]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <thead> <th></th> <th>Age</th> <th>Tumor_Size</th> </thead>
<tbody> <th>count</th> <th>mean</th> <th>std</th> <th>min</th>
<th>25%</th> <th>50%</th> <th>75%</th> <th>max</th> </tbody>
<tbody></tbody>
```

5.000000	5.000000
----------	----------

49.000000	2.900000
-----------	----------

8.514693	0.987421
----------	----------

38.000000	1.800000
-----------	----------

45.000000	2.100000
-----------	----------

49.000000	2.900000
-----------	----------

52.000000	3.500000
-----------	----------

61.000000	4.200000
-----------	----------

In []:

In []: